



legoESM Scientific Guide

A Fully Differentiable Earth System Model in JAX

Version 4.0 — March 2026

legoESM Development Team

This document provides a comprehensive scientific description of the legoESM Earth System Model, including governing equations, numerical methods, and parameterisation schemes for all model components. The structure follows the NCAR CAM Scientific Guide convention of self-contained chapters for dynamics, physics, ocean, land, and sea ice.

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Chapter 1

Introduction

1.1 Overview

legoESM is a modular, fully differentiable Earth System Model (ESM) written entirely in JAX. Every component—atmosphere, ocean, land, sea ice, lake, and biogeochemistry—is compatible with JAX’s automatic differentiation (`jit`, `grad`, `vmap`, `scan`), enabling novel applications in data assimilation (4D-Var with exact adjoints), parameter estimation, hybrid AI–physics modelling, and sensitivity analysis.

The model features:

- **25+ dynamical cores** spanning five discretisation families (C-D grid FV3, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi), with shallow water, hydrostatic primitive equation, and non-hydrostatic compressible Euler equation sets;
- **35+ physics parameterisations** for radiation, convection, microphysics, boundary-layer turbulence, clouds, gravity wave drag, and surface exchange;
- A **3D Boussinesq ocean** with split-explicit barotropic solver, KPP vertical mixing, GM-Redi eddy parameterisation, and biogeochemistry (abiotic carbon cycle + NPZD ecosystem);
- A **multi-layer land model** with Richards-equation soil hydrology, Johansen thermal diffusion, Farquhar–Ball–Berry photosynthesis, and a DALEC-990 carbon cycle;
- **Sea-ice dynamics** with EVP rheology, multi-category thickness distribution, and thermodynamic coupling;
- A **tile-based coupler** with Monin–Obukhov surface layer, area-weighted flux blending, and radiation sub-cycling;
- **CMIP6-class infrastructure**: CF/CMOR output, six experiment templates, checkpointing with SHA-256 reproducibility verification, and a 16-parameter tuning registry;
- A **Spherical Fourier Neural Operator** (SFNO) for learned dynamics and physics emulation.

1.2 Design Principles

1. **JAX-first**: All code uses pure-functional JAX with no Python loops in hot paths. Conditional branching uses `jnp.where` or `jax.lax.cond` to preserve differentiability.
2. **Grid-agnostic physics**: Operators implement a shared protocol (`GridProtocol`); physics schemes accept any grid via column adapters that reshape to `(ncol, nlev)`.
3. **Modular composition**: Each parameterisation follows a factory pattern returning a callable `physics_fn(state, grid, coord) → Tendencies`.

4. **Unified C-D grid:** Atmosphere and ocean share the same FV3-style C-D grid operator module on the cubed sphere, eliminating code duplication and the Hollingsworth–Kallberg instability.
5. **Pytree registration:** Field and state objects are registered as JAX pytrees for seamless autodiff through the entire model.

1.3 Notation and Conventions

Throughout this document:

- Vectors are denoted in bold: $\mathbf{v} = (u, v)$;
- $\hat{f}$ denotes spectral (spherical harmonic) coefficients;
- Subscript k indexes vertical levels (increasing downward);
- Half-integer subscripts ($k \pm \frac{1}{2}$) denote layer interfaces;
- SI units are used unless otherwise noted;
- $f = 2\Omega \sin \phi$ is the Coriolis parameter.

Chapter 2

Grid Systems and Coordinates

2.1 Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection**. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be gnomonic coordinates on a face.

2.1.1 Face-to-Cartesian Mapping

For face 0 (+x face):

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y). \quad (2.1)$$

Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, with geographic coordinates $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

2.1.2 Cell Area

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (2.2)$$

2.1.3 Grid Spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (2.3)$$

The minimum grid spacing on a C_n cubed sphere is

$$\Delta x_{\min} = \frac{\pi R}{2n\sqrt{3}}, \quad (2.4)$$

occurring at the cube corners where the gnomonic distortion is maximal.

2.1.4 Grid Angle and Wind Rotation

The angle θ between the grid x -axis and geographic east:

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (2.5)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (2.6)$$

2.1.5 Halo Exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back. Two halo widths are supported: `halo=1` for second-order and MUSCL stencils, and `halo=2` for PPM/WENO5 and biharmonic operators.

Native 4D halo exchange. All 3D operators use a 4D halo path (`pad_halo_4d`, `pad_halo_vector_4d`) that exchanges all vertical levels in a single MPI message per neighbour, replacing the earlier `vmap(pad_halo)` path which issued n_{lev} separate messages.

MPI differentiability. All `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp` in `halo_exchange.py`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Only `allreduce(SUM)` is AD-safe among MPI reductions; `MAX/MIN/allgather/bcast` are used only in diagnostics and are not differentiable.

2.2 Gaussian Grid and Spectral Transforms

The spectral dynamical cores use a Gaussian grid with triangular truncation at total wavenumber n_{max} .

2.2.1 Grid Dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\text{max}} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\text{max}} + 1)(n_{\text{max}} + 2)}{2}. \quad (2.7)$$

Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced.

2.2.2 Associated Legendre Polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. Computed via the recursion:

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (2.8)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (2.9)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

2.2.3 Spherical Harmonic Transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (2.10)$$

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\max}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (2.11)$$

2.2.4 Spectral Operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

2.2.5 Velocity from Vorticity and Divergence

Streamfunction $\psi = \nabla^{-2} \zeta$, velocity potential $\chi = \nabla^{-2} \delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (2.12)$$

where θ is colatitude.

2.3 Lat-Lon Grid

A uniform $n_{\text{lat}} \times n_{\text{lon}}$ latitude-longitude grid with A-grid (collocated) staggering. All operators use second-order centred differences. A Fourier polar filter is applied to tendencies at high latitudes to prevent the CFL constraint from becoming prohibitively small near the poles.

2.4 Voronoi/MPAS Grid

An unstructured Voronoi tessellation generated from an icosahedral parent mesh. The TRiSK (Thuburn–Ringler–Skamarock–Klemp) scheme provides mimetic discrete operators on this grid:

- Scalars (height, tracers, pressure) at cell centres;
- Normal velocity at edges;
- Vorticity at vertices (triangle dual).

Variable resolution is supported by mesh refinement.

2.5 Vertical Coordinates

2.5.1 Sigma Coordinate

$$\sigma = p / p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (2.13)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$.

Simmons–Burridge Geopotential

The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (2.14)$$

where the Simmons–Burridge coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta\sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (2.15)$$

Sigma-Dot (Vertical Velocity)

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}. \quad (2.16)$$

Pressure Velocity

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}. \quad (2.17)$$

2.5.2 Hybrid Sigma-Pressure Coordinate

$$p(k) = A(k) p_{\text{ref}} + B(k) p_s, \quad (2.18)$$

where $A(k) \rightarrow 1$, $B(k) \rightarrow 0$ at the model top (pure pressure levels) and $A(k) \rightarrow 0$, $B(k) \rightarrow 1$ near the surface (pure sigma). This eliminates large pressure-gradient force errors over steep topography that afflict the pure sigma coordinate.

2.5.3 Terrain-Following Height Coordinate (z^*)

Used by the non-hydrostatic compressible Euler equations:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (2.19)$$

A 1D reference state $(\rho_0(z), \theta_0(z), \pi_0(z))$ is pre-computed for the perturbation formulation.

2.5.4 Ocean z^* Coordinate

The ocean uses a dynamic z^* coordinate with free-surface elevation η :

$$J(\eta) = \frac{\eta + H}{H_{\text{max}}}, \quad (2.20)$$

where H is the resting depth.

Chapter 3

Coupling of Dynamics and Physics

3.1 Physics Pipeline

The `PhysicsPipeline` class composes all physical parameterisations into a single callable that is invoked once per dynamics timestep. The calling sequence within a single step is:

1. **Dynamics step:** Update winds, temperature, surface pressure, and tracers via the selected dynamical core.
2. **Physics step:** Apply parameterisations in the following order:
 - (a) Radiation (with optional sub-cycling),
 - (b) Deep convection,
 - (c) Shallow convection / large-scale condensation,
 - (d) Cloud microphysics,
 - (e) Boundary-layer turbulence and vertical diffusion,
 - (f) Gravity wave drag,
 - (g) Surface fluxes (tile coupler).
3. **Conservation fixers:** Mass, moisture, and optionally energy.
4. **Diagnostics:** Energy budget, monthly means, snapshots.

3.2 Radiation Sub-Cycling

Radiation is the most expensive physics component. The pipeline supports calling radiation every N_{rad} dynamics steps while holding radiative heating rates constant between calls. The branching is implemented via `jax.lax.cond` with both branches returning identical array shapes and dtypes.

3.3 Grid-Agnostic Column Adapter

The `ColumnAdapter` reshapes any grid’s 3D state into a `(ncol, nlev)` column format for physics, then unflattens back to the native grid shape. This allows all physics schemes to operate on simple column arrays regardless of the underlying horizontal discretisation.

3.4 Time-Split vs Process-Split

legoESM uses **time-split** coupling: dynamics and physics operate sequentially within each timestep. Physics tendencies are computed from the dynamics-updated state and applied as a forward-Euler

update. This is natural for the finite-volume dycores that use a Lagrangian vertical coordinate internally.

Chapter 4

Atmospheric Dynamics

This chapter describes the governing equations and numerical methods for each dynamical core. Cores are organised by equation system (shallow water, hydrostatic primitive equations, non-hydrostatic compressible Euler), then by discretisation.

4.1 Shallow Water Equations

The rotating shallow water equations in **vector-invariant form**:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (4.1)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (4.2)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (4.3)$$

where $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ is the Bernoulli function, $\zeta = \partial v / \partial x - \partial u / \partial y$ is the relative vorticity, and D_u, D_v represent hyperdiffusion.

4.1.1 C-D Grid Cubed-Sphere Discretisation

This is the primary shallow water solver, using the FV3-style C-D grid staggering of [Lin \(2004\)](#) and [Putman and Lin \(2007\)](#).

Prognostic variables. D-grid winds (u_d, v_d) at cell corners, shape $(6, n+1, n+1)$; height h at cell centres, shape $(6, n, n)$.

D-grid vorticity. Vorticity at each cell corner is the discrete circulation divided by the corner dual-cell area:

$$\zeta_{\text{corner}} = \frac{1}{A_c} \oint \mathbf{u} \cdot d\mathbf{l} \approx \frac{1}{A_c} [u_{i,j}^D \Delta x - u_{i+1,j}^D \Delta x + v_{i,j+1}^D \Delta y - v_{i,j}^D \Delta y]. \quad (4.4)$$

This circulation-based discretisation is exact on the D-grid and **eliminates the Hollingsworth–Kallberg instability** that plagues A-grid vorticity operators.

Arakawa–Lamb gradient. The Bernoulli gradient at D-grid corners uses a 4-point average:

$$\left. \frac{\partial B}{\partial x} \right|_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{B_{i+1,j} + B_{i+1,j+1} - B_{i,j} - B_{i,j+1}}{2\Delta x}. \quad (4.5)$$

C-grid mass flux. C-grid velocities are diagnosed from D-grid winds by face-edge averaging. Scalar transport uses upwind C-grid mass fluxes:

$$\frac{dh}{dt} = -\frac{1}{A} [(hu_c)_{i+\frac{1}{2},j} - (hu_c)_{i-\frac{1}{2},j} + (hv_c)_{i,j+\frac{1}{2}} - (hv_c)_{i,j-\frac{1}{2}}]. \quad (4.6)$$

FV3-faithful PPM transport (v3.9). As of v3.9, the C-D grid operator module underwent a complete rewrite to implement FV3-faithful Piecewise Parabolic Method (PPM) transport (Colella and Woodward, 1984) on staggered D/C grids. Key elements include:

- PPM reconstruction with Colella–Woodward monotonicity limiters for all scalar fields;
- Flux-form mass transport on staggered D-grid and C-grid;
- Corrected geographic-to-grid wind rotation eliminating sign errors in cross-face interpolation;
- Tuned hyperdiffusion with physical e -folding time ($\nu_4 = \Delta x^4 / \tau_{\text{efold}}$).

The module spans approximately 740 lines of JAX code.

FV3 forward–backward shallow-water core (v4.0). An FV3-exact implementation of the complete `c_sw/d_sw` forward–backward stepping algorithm is provided in `fv3_sw_core.py`. The forward step (`c_sw`) computes C-grid mass fluxes and height tendencies; the backward step (`d_sw`) updates D-grid winds using the vorticity flux and Bernoulli gradient.

d2a2c_vect. The D-grid $\rightarrow$ A-grid $\rightarrow$ C-grid vector conversion faithfully ports the GFDL FV3 algorithm, including:

- Edge-boundary stencils with one-sided 4-point interpolation at cube-face edges;
- Non-orthogonality corrections via $\cos \alpha$ and $(\sin^2 \alpha)^{-1}$ corner metrics;
- Halo-aware padding with face-boundary detection.

PPM wind and vorticity transport. C-grid wind tendencies (`xtp_u`, `ytp_v`) and absolute vorticity flux use Piecewise Parabolic Method reconstruction with FV3-specific interpolation coefficients (A_1 , A_2 , C_1 – C_3) and Colella–Woodward monotonicity limiters.

Cell-centred kinetic energy. Kinetic energy is computed from A-grid velocities at cell centres, with an optional 4th-order averaging stencil (`a2b_ord4`).

Duo-grid sub-grid metrics (Mouallem et al. 2023). The cubed-sphere C-D grid computes 9-point sub-grid angle metrics per cell:

$$\sin \alpha_{\text{sg}}, \cos \alpha_{\text{sg}} \in \mathbb{R}^{6 \times n \times n \times 9}, \quad (4.7)$$

encoding the local deviation from orthogonality at cell centres, edge midpoints, and corners. These enable FV3-faithful flux scaling at cube-face boundaries where the gnomonic projection introduces non-orthogonality. Additional C-D grid metrics:

- Centre-to-centre distances Δx_c (shape $6 \times (n+1) \times n$) and Δy_c (shape $6 \times n \times (n+1)$);
- Non-orthogonality factors $\cos \alpha_{\text{corner}}$ and $(\sin^2 \alpha_{\text{corner}})^{-1}$ at D-grid corners.

4.1.2 Lat-Lon Finite Volume Discretisation

A-grid staggering on a uniform $n_{\text{lat}} \times n_{\text{lon}}$ grid. Centred differences for all operators. A Fourier polar filter damps meridional wavelengths near the poles to maintain CFL stability.

4.1.3 C-Grid Lat-Lon Discretisation

Height at cell centres; u at longitude interfaces; v at latitude interfaces. Advantages: exact pressure gradient (just cell differences), mass flux naturally at cell interfaces (no interpolation), and better gravity wave dispersion. The Coriolis term requires 4-point interpolation.

4.1.4 TRiSK Discretisation on Voronoi Mesh

Following [Ringler et al. \(2010\)](#), the discrete operators on the Voronoi mesh are:

$$\text{Divergence at cell } c : (\nabla \cdot \mathbf{u})_c = \frac{1}{A_c} \sum_e n_{e,c} u_e l_e, \quad (4.8)$$

$$\text{Gradient at edge } e : (\nabla \phi)_e = \frac{\phi_{c_2} - \phi_{c_1}}{d_e}, \quad (4.9)$$

$$\text{Curl at vertex } v : (\nabla \times \mathbf{u})_v = \frac{1}{A_v} \sum_e t_{e,v} u_e d_e, \quad (4.10)$$

where $n_{e,c}$ and $t_{e,v}$ are orientation signs, l_e is edge length, d_e is distance between adjacent cell centres, and A_c, A_v are cell/vertex areas.

4.1.5 Spectral Shallow Water

Vorticity–divergence formulation on the Gaussian grid:

$$\frac{\partial \hat{\zeta}}{\partial t} = -\nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}], \quad (4.11)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (E + \widehat{\Phi} + \Phi_s), \quad (4.12)$$

$$\frac{\partial \hat{\Phi}}{\partial t} = -\nabla \cdot (\widehat{\Phi} \mathbf{v}), \quad (4.13)$$

where $E = \frac{1}{2}(u^2 + v^2)$ and $\Phi = gh$. Pseudospectral: nonlinear products computed on the grid, then transformed to spectral space.

4.2 Hydrostatic Primitive Equations

Prognostic variables: u, v (horizontal winds), T (temperature), p_s (surface pressure), and tracers q_i .

4.2.1 Continuous Equations

In σ -coordinates:

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (4.14)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (4.15)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (4.16)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (4.17)$$

where $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, and $\kappa = R_d/c_p$. The adiabatic heating term $\kappa T \omega/p$ represents compression warming during descent.

4.2.2 C-D Grid Cubed-Sphere Discretisation

Prognostic state: u_d, v_d at D-grid corners $(6, n+1, n+1, N_{\text{lev}})$; T and p_s at cell centres. Spatial operators are identical to the shallow water C-D grid (Section 4.1), extended to 3D with:

- Simmons–Burridge geopotential integration (Section 2.5);
- Sigma-dot computation with discrete continuity closure;
- Upwind vertical advection: $-\dot{\sigma} \partial f / \partial \sigma$.

4.2.3 Spectral Transform Discretisation

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972):

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\zeta}, \quad (4.18)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (K + \Phi + \widehat{R_d T \ln p_s}) + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\delta}, \quad (4.19)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T \mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \widehat{\partial T / \partial \sigma} + \widehat{\kappa T \omega / p}, \quad (4.20)$$

$$\frac{\partial \widehat{\ln p_s}}{\partial t} = - \int_0^1 \hat{\delta} d\sigma. \quad (4.21)$$

Spectral curl and divergence. For flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$:

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (4.22)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices. The $\cos \phi$ weighting is absorbed into the Legendre basis, making the formulation pole-safe.

Semi-implicit treatment of gravity waves. Following Hoskins and Simmons (1975), the fast gravity-wave terms are treated implicitly. Linearising about a reference temperature $T_{\text{ref}} = 300$ K yields a Helmholtz equation per spectral mode:

$$[\mathbf{I} + \alpha^2 \Delta t^2 \lambda_n \mathbf{\Gamma}] \hat{\delta}^{n+1} = \hat{\delta}^{\text{explicit}}, \quad (4.23)$$

where $\lambda_n = n(n+1)/a^2$ is the Laplacian eigenvalue and $\mathbf{\Gamma}$ is the vertical coupling matrix. This is pre-factored (LU decomposition) for each wavenumber n at initialisation, removing the gravity-wave CFL constraint and allowing 3–4 $\times$ larger timesteps.

Verification (v3.9). A comprehensive spectral dycore test suite validates conservation (mass, energy, enstrophy), long-integration stability under Held–Suarez forcing, spectral convergence with increasing truncation wavenumber, and semi-implicit scheme correctness. See Chapter 17 for details.

4.2.4 Lat-Lon Finite-Volume Discretisation

Identical equations to the C-D grid version, discretised on the A-grid lat-lon mesh with centred differences. Polar filter applied to all tendencies. Optional Rayleigh sponge layer above a configurable σ cutoff.

4.2.5 MPAS/Voronoi Discretisation

TRiSK operators (Section 4.1) extended to 3D. Uses RK4 or SSP-RK3 time stepping with energy- or enstrophy-conserving potential vorticity flux options.

4.3 Non-Hydrostatic Compressible Euler Equations

In the z^* terrain-following coordinate, the prognostic variables are perturbations from a hydrostatically balanced reference state $(\rho_0, \theta_0, \pi_0)$:

$$\frac{\partial u}{\partial t} = (\zeta + f)v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (4.24)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f)u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (4.25)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (4.26)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (4.27)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (4.28)$$

where $r(z)$ is the Rayleigh sponge coefficient and the Exner function perturbation is

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (4.29)$$

4.3.1 Split-Explicit Time Integration

Slow modes (SSP-RK3): advection, Coriolis, horizontal pressure gradient force, and diffusion.

Fast modes (forward–backward acoustic substeps): vertical sound/gravity waves. Typically $N_s = 6$ substeps per RK stage.

Forward step (vertical velocity).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (4.30)$$

Backward step (density, potential temperature).

$$\rho'^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta'^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (4.31)$$

Acoustic off-centering. To damp vertically propagating acoustic/gravity wave noise ([Skamarock and Klemp, 2008](#)):

$$\rho_\beta'^{n+1} = (1 + \beta) \rho'^{n+1} - \beta \rho'^n, \quad (4.32)$$

where $\beta \geq 0$ is the off-centering parameter (default 0.1).

4.3.2 Sponge Layer

$$r(z) = r_{\max} \sin^2 \left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}} \right). \quad (4.33)$$

4.4 Finite-Volume Transport

4.4.1 MUSCL Reconstruction (2nd Order)

Piecewise-linear reconstruction with slope limiting ([van Leer, 1979](#)):

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (4.34)$$

with minmod or MC (monotonised central) limiters.

4.4.2 PPM Reconstruction (4th Order)

The Piecewise Parabolic Method ([Colella and Woodward, 1984](#)) constructs a monotone parabola in each cell. Fourth-order edge values:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (4.35)$$

Monotonicity constraint: $\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1})$. Colella–Woodward limiting prevents new extrema in the parabolic profile.

4.4.3 WENO5 Reconstruction (5th Order)

Three sub-stencil polynomials combined with nonlinear weights adapting to local smoothness ([Jiang and Shu, 1996](#)):

$$q_L = \sum_k \omega_k q^{(k)}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad \alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad (4.36)$$

where d_k are ideal weights (1/10, 6/10, 3/10) and β_k are smoothness indicators.

4.4.4 Rusanov Flux

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (4.37)$$

where $c = \sqrt{g h}$ is the gravity-wave speed.

4.5 Time Integration

4.5.1 SSP-RK3

The three-stage strong-stability-preserving Runge–Kutta method (Shu and Osher, 1988):

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (4.38)$$

$$\mathbf{u}^{(2)} = \frac{3}{4}\mathbf{u}^n + \frac{1}{4}\mathbf{u}^{(1)} + \frac{1}{4}\Delta t F(\mathbf{u}^{(1)}), \quad (4.39)$$

$$\mathbf{u}^{n+1} = \frac{1}{3}\mathbf{u}^n + \frac{2}{3}\mathbf{u}^{(2)} + \frac{2}{3}\Delta t F(\mathbf{u}^{(2)}). \quad (4.40)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

4.5.2 Robert–Asselin–Williams (RAW) Time Filter

The leapfrog scheme used by the semi-implicit spectral PE solver requires a time filter to damp the computational mode. The Williams (2009) modification:

$$d^n = \frac{\gamma}{2}(X^{n-1} - 2X^n + X^{n+1}), \quad (4.41)$$

$$X_{\text{filt}}^n = X^n + (1 - \alpha) d^n, \quad (4.42)$$

$$X_{\text{filt}}^{n+1} = X^{n+1} + \alpha d^n, \quad (4.43)$$

with $\alpha = 0.5$ (default). This restores second-order accuracy while preserving the damping of the computational mode.

4.5.3 CFL Conditions and Adaptive Hyperdiffusion

The CFL-limited timestep:

$$\Delta t_{\text{max}} = \frac{C_{\text{CFL}} \Delta x_{\text{min}}}{(u_{\text{max}} + c_{\text{gw}}) \sqrt{n_{\text{dim}}}}, \quad (4.44)$$

where $c_{\text{gw}} = \sqrt{g H}$ is the gravity wave speed.

The maximum stable hyperdiffusion coefficient for p -th order:

$$\nu_p = S \cdot \frac{C(p) \Delta x^{2p}}{\Delta t}, \quad C(2) = \frac{1}{2}, \quad C(4) = \frac{1}{8}, \quad C(6) = \frac{1}{48}, \quad (4.45)$$

where $S \in (0, 1]$ is a safety factor. The physical e-folding time for $2\Delta x$ waves is $\tau_{\text{efold}} = \Delta x^{2p} / \nu_p$.

Chapter 5

Atmospheric Physics

5.1 Thermodynamic Preliminaries

5.1.1 Saturation Mixing Ratio

Using the Tetens (August–Roche–Magnus) formula:

$$e_s(T) = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d/R_v \approx 0.622, \quad (5.1)$$

where $T_c = T - 273.15$ K. Over ice, the Clausius–Clapeyron form is used:

$$e_{s,i}(T) = 611.2 \exp\left[\frac{L_s}{R_v} \left(\frac{1}{T_{\text{freeze}}} - \frac{1}{T}\right)\right]. \quad (5.2)$$

5.1.2 Virtual Temperature

$$T_v = T (1 + 0.608 q_v - q_c - q_r - q_i). \quad (5.3)$$

5.1.3 Moist Adiabatic Lapse Rate

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}. \quad (5.4)$$

5.2 Radiation

5.2.1 Gray Two-Stream Radiation

Following [Frierson et al. \(2006\)](#):

Longwave Optical Depth

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \sum_k \tau_{\text{moist}} q_{v,k} \frac{\Delta p_k}{g}, \quad (5.5)$$

where the moisture optical depth per layer is $\Delta \tau_{\text{moist},k} = \tau_{\text{moist}} q_{v,k} \Delta p_k / g$, with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 0.0115 \text{ m}^2 \text{ kg}^{-1}$.

Two-Stream Longwave Fluxes

Layer transmittance and emissivity:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (5.6)$$

The diffusivity factor $D \approx 5/3$ accounts for the hemispheric-mean angular integration.

Upward sweep from surface ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$):

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (5.7)$$

Downward sweep from TOA ($F_0^\downarrow = 0$):

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (5.8)$$

Shortwave (Beer–Lambert)

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-\tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}), \quad (5.9)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1361 \text{ W m}^{-2}$, $\alpha_s = 0.31$. Only the downward beam is absorbed; reflected SW escapes to space.

Heating Rate

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^\uparrow - F_{\text{net},k-1/2}^\uparrow}{p_{k+1/2} - p_{k-1/2}}. \quad (5.10)$$

5.2.2 RRTMGP Correlated- k Radiation

The RRTMGP backend uses JAX-RRTMGP with:

- 128 LW and 112 SW g -points for correlated- k gas absorption;
- Two-stream delta-Eddington radiative transfer equation;
- Spectral line absorption for H_2O , CO_2 , CH_4 , N_2O , O_3 ;
- Cloud optics via liquid/ice water path and effective radii lookup tables;
- Optional aerosol optical depth with prescribed single-scattering albedo ($\omega_0 = 0.93$) and asymmetry factor ($g = 0.70$).

Default trace-gas concentrations: $\text{CO}_2=415 \text{ ppmv}$, $\text{CH}_4=1900 \text{ ppbv}$, $\text{N}_2\text{O}=332 \text{ ppbv}$.

5.2.3 Solar Geometry

The cosine of the solar zenith angle:

$$\mu_0 = \sin \phi \sin \delta + \cos \phi \cos \delta \cos h, \quad (5.11)$$

where δ is the solar declination and h is the hour angle. Daily-mean insolation and perpetual-equinox modes are also available.

5.3 Deep Convection

Five convection schemes are available, all fully JAX-differentiable.

5.3.1 Simplified Betts–Miller (SBM) Relaxation

Following Frierson (2007):

CAPE Computation

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (5.12)$$

Smooth Trigger

$$\mathcal{F} = \sigma(\kappa_s(\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (5.13)$$

Enthalpy-Conserving Relaxation

The reference profile T_{ref} is adjusted so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p(T_{\text{ref},k} - T_k) + L_v(q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (5.14)$$

Tendencies:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (5.15)$$

Precipitation

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (5.16)$$

5.3.2 Kuo Column Moisture-Excess Scheme

Convection proportional to column-integrated moisture excess above saturation $[\text{kg m}^{-2}]$. Heating and moistening are partitioned via a parameter α_{heat} :

$$MC = \int \max(q_v - q_s, 0) \frac{dp}{g}, \quad (5.17)$$

$$\frac{dT}{dt} = \mathcal{F} \alpha_{\text{heat}} \frac{T_{\text{moist}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} (1 - \alpha_{\text{heat}}) \frac{q_{s,\text{moist}} - q_v}{\tau_c}, \quad (5.18)$$

with $\alpha_{\text{heat}} = 0.75$ and $\tau_c = 3600 \text{ s}$.

5.3.3 Mass-Flux Convection (Arakawa–Wu Type)

A prognostic base mass flux M_c relaxes toward an equilibrium value:

$$M_{\text{eq}} = \sigma[(\text{CAPE} - \text{CAPE}_{\text{thresh}})/\text{scale}] \cdot M_{\text{scale}}, \quad \frac{dM_c}{dt} = \frac{M_{\text{eq}} - M_c}{\tau_{\text{adj}}}. \quad (5.19)$$

The mass-flux profile is sinusoidal: $m(p) = \sin[\pi(p_{\text{base}} - p)/\Delta p]$. The entraining updraft uses exponential dilution:

$$T_u(z) = e^{-\epsilon_0 z} T_{\text{moist}} + (1 - e^{-\epsilon_0 z}) T_{\text{env}}, \quad (5.20)$$

with entrainment rate $\epsilon_0 = 10^{-3} \text{ m}^{-1}$ and detrainment rate $\delta_0 = 10^{-3} \text{ m}^{-1}$.

Precipitation. Precipitation is diagnosed from detrainment of updraft condensate, integrated over the column using layer thickness Δz :

$$P = \sum_k \delta_0 M_k q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{s}^{-1}], \quad (5.21)$$

where $q_{\ell,k} = \max(q_{u,k} - q_{s,k}, 0)$ is the updraft condensate and $M_k = M_c m(p_k)$ is the mass flux profile $[\text{kg m}^{-2} \text{s}^{-1}]$. The vertical measure Δz is used (not $\Delta p/g$) because M already carries units of mass flux per unit area, so $\delta_0 M q_\ell$ has units $[\text{kg m}^{-3} \text{s}^{-1}]$ and multiplying by Δz [m] yields $[\text{kg m}^{-2} \text{s}^{-1}]$.

5.3.4 EDMF Convection

Eddy-Diffusivity Mass-Flux with a prognostic updraft area fraction a_u :

$$\frac{da_u}{dt} = \frac{a_{u,\text{eq}} - a_u}{\tau_a}, \quad (5.22)$$

combining mass-flux (updraft) and eddy-diffusivity (environmental mixing) contributions.

Precipitation. As in the mass-flux scheme (§5.4), precipitation is diagnosed from condensate detrainment using Δz :

$$P = \sum_k \delta_0 M_{u,k} q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{s}^{-1}], \quad (5.23)$$

with $M_{u,k} = \rho_k a_u w_{u,k}$ and $\delta_0 = 2 \times 10^{-3} \text{ m}^{-1}$.

5.3.5 Deep Convective Adjustment

The simplest parameterisation: scans bottom-to-top adjusting unstable layer pairs toward moist-adiabatic neutrality while conserving total enthalpy. Uses sigmoid activation for smooth, differentiable blending.

5.4 Cloud Microphysics

5.4.1 Kessler Warm-Rain Scheme

Prognostic variables: q_c (cloud water), q_r (rain).

Saturation adjustment. The saturation excess is converted to a tendency (rate) by dividing by the time step Δt :

$$\left. \frac{dq_v}{dt} \right|_{\text{sat}} = - \frac{\sigma(\kappa_s(q_v - q_s)) \cdot (q_v - q_s)}{\Delta t}, \quad (5.24)$$

with sigmoid sharpness κ_s . This ensures all microphysics outputs have consistent units of $\text{kg kg}^{-1} \text{s}^{-1}$.

Autoconversion (cloud $\rightarrow$ rain).

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{auto}} \text{softplus}(q_c - q_{c,\text{thresh}}), \quad k_{\text{auto}} = 10^{-3} \text{ s}^{-1}, \quad q_{c,\text{thresh}} = 10^{-3} \text{ kg/kg}. \quad (5.25)$$

Accretion (collection of cloud by rain).

$$\left. \frac{dq_c}{dt} \right|_{\text{accr}} = -k_{\text{accr}} q_c q_r^{0.875}, \quad k_{\text{accr}} = 2.2. \quad (5.26)$$

Rain evaporation.

$$\left. \frac{dq_r}{dt} \right|_{\text{evap}} = k_{\text{evap}} \frac{q_s - q_v}{q_s} q_r^{0.525}. \quad (5.27)$$

Sedimentation. Fall speed: $V_t = V_0 \sqrt{\rho_{\text{sfc}}/\rho}$ with $V_0 = 5$ m/s.

Latent heating. $dT/dt = (L_v/c_p) \cdot (\text{condensation} - \text{evaporation})$.

5.4.2 Seifert–Beheng Two-Moment Warm Rain

Prognostic mass (q_c, q_r) and number concentration (N_c, N_r).

Autoconversion.

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{au}} q_c^2 \sigma[\kappa(x_c - x^*)], \quad x_c = \frac{q_c \rho}{N_c}, \quad (5.28)$$

where $k_{\text{au}} = 600 \text{ m}^3/(\text{kg s})$, $x^* = 2.6 \times 10^{-10} \text{ kg}$ is the separation mass, and the sigmoid onset replaces the hard threshold.

Self-collection and breakup.

$$\left. \frac{dN_r}{dt} \right|_{\text{sc}} = -k_{\text{sc}} N_r q_r \rho, \quad \left. \frac{dN_r}{dt} \right|_{\text{br}} = +k_{\text{br}} \sigma[\kappa(D_r - D_{\text{eq}})] \left| \left. \frac{dN_r}{dt} \right|_{\text{sc}} \right|. \quad (5.29)$$

5.4.3 Morrison Double-Moment (Ice + Liquid)

Extends the warm-rain scheme with ice-phase processes:

Ice nucleation (Cooper 1986).

$$N_{i,\text{target}} = \frac{N_{i,0}}{\rho} \exp[a_C(T_{\text{freeze}} - T)]. \quad (5.30)$$

Depositional growth.

$$\left. \frac{dq_i}{dt} \right|_{\text{dep}} = k_{\text{dep}} \text{softplus}(S_i) \sqrt{N_i}, \quad S_i = \frac{q_v}{q_{s,i}} - 1. \quad (5.31)$$

Bergeron process (mixed-phase). Cloud water converts to ice in the temperature window $T_{\text{center}} - T_{\text{width}} < T < T_{\text{freeze}}$:

$$\left. \frac{dq_c}{dt} \right|_{\text{Berg}} = -k_{\text{Berg}} q_c f_{\text{ice}}(T). \quad (5.32)$$

Riming, aggregation, melting. Riming: $q_c \rightarrow q_i$ and $q_c \rightarrow q_s$ via collection. Aggregation: $q_i \rightarrow q_s$ when crystals grow large enough. Melting: sigmoid-activated near T_{freeze} .

Fall speeds (power law).

$$V_t = a \left(\frac{\rho q}{\rho_{\text{sfc}}} \right)^b, \quad \text{rain: } a = 130, b = 0.5; \quad \text{ice: } a = 50, b = 0.25; \quad \text{snow: } a = 30, b = 0.3. \quad (5.33)$$

5.4.4 Thompson Hybrid-Moment Microphysics

Extends Morrison with graupel. Rimed ice exceeding a critical threshold converts to graupel:

$$\left. \frac{dq_g}{dt} \right|_{\text{rime}} = f_g \left. \frac{dq_i}{dt} \right|_{\text{rime}} \cdot \sigma \left[\kappa \left(\left. \frac{dq_i}{dt} \right|_{\text{rime}} - R_{\text{thresh}} \right) \right]. \quad (5.34)$$

5.4.5 Summary of Microphysics Schemes

Scheme	Moments	Species	Key features
Kessler	1	q_c, q_r	Warm rain only
Seifert–Beheng	2	q_c, q_r, N_c, N_r	Realistic droplet spectra
Morrison	2	$q_c, q_r, q_i, q_s, N_c, N_r, N_i$	Full mixed-phase
Thompson	Hybrid	$+q_g$	Graupel, riming

5.5 Cloud Fraction

5.5.1 Sundqvist (1988)

Diagnostic cloud fraction based on a relative humidity threshold:

$$c_f = \text{clamp} \left(\frac{\text{RH} - \text{RH}_{\text{crit}}}{1 - \text{RH}_{\text{crit}}}, 0, 1 \right), \quad \text{RH}_{\text{crit}} = 0.8. \quad (5.35)$$

5.5.2 Xu–Randall (1996)

Combines relative humidity and condensate amount:

$$c_f = \text{RHP} \left[1 - \exp \left(- \frac{\alpha q_c}{(1 - \text{RH}) q_s} \right) \right], \quad p = 2, \alpha = 100. \quad (5.36)$$

5.6 Boundary-Layer Turbulence

5.6.1 Louis (1979) Local Closure

Mixing Length

$$l = \frac{\kappa_{\text{vK}} z}{1 + \kappa_{\text{vK}} z / l_{\text{max}}}, \quad \kappa_{\text{vK}} = 0.4, \quad l_{\text{max}} = 300 \text{ m}. \quad (5.37)$$

Richardson Number

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2. \quad (5.38)$$

Stability Functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{ Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{ Ri} / \sqrt{1 + d \text{ Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (5.39)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

5.6.2 Holtslag–Boville Nonlocal K-Profile

PBL Height Diagnosis

Bulk Richardson number:

$$\text{Ri}_{\text{bulk}}(z) = \frac{g}{\theta_{v,\text{sfc}}} \frac{(\theta_v - \theta_{v,\text{sfc}}) z}{|\mathbf{V}|^2}. \quad (5.40)$$

PBL height h is diagnosed where Ri_{bulk} crosses $\text{Ri}_{\text{crit}} = 0.25$, using sigmoid-weighted averaging for a smooth, differentiable diagnosis.

K-Profile Inside PBL

$$K_m = \kappa_{\text{vK}} u_* z (1 - z/h)^2. \quad (5.41)$$

Above the PBL, Louis (1979) Ri-dependent mixing applies.

5.6.3 TKE-Based Closure

Prognostic turbulent kinetic energy:

$$\frac{\partial \text{TKE}}{\partial t} = P - \varepsilon + D + B, \quad (5.42)$$

where P is shear production, $\varepsilon = c_\varepsilon \text{TKE}^{3/2}/l$ is dissipation, D is diffusion, and B is buoyancy production/destruction. Closure: $K_m = c_m \sqrt{\text{TKE}} \cdot l$, $K_h = K_m / \text{Pr}_t$.

5.6.4 Implicit Vertical Diffusion

The backward-Euler tridiagonal system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm:

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (5.43)$$

5.7 Surface Exchange Formulations

5.7.1 Neutral Bulk Aerodynamic Fluxes

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (5.44)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad \text{LH} = \rho L_v C_E |\mathbf{V}| (q_s - q_a), \quad (5.45)$$

with $C_D = C_H = C_E = 1.5 \times 10^{-3}$.

5.7.2 Monin–Obukhov Similarity Theory (MOST)

Obukhov length.

$$L = -\frac{u_*^3 T_v}{\kappa_{\text{vK}} g (\theta_* + 0.61 T_a q_*)}. \quad (5.46)$$

Businger–Dyer stability functions.

$$\psi_m(\zeta) = \begin{cases} 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan x + \frac{\pi}{2}, & \zeta < 0, \quad x = (1 - 16\zeta)^{1/4}, \\ -5\zeta, & \zeta \geq 0, \end{cases} \quad (5.47)$$

$$\psi_h(\zeta) = \begin{cases} 2 \ln\left(\frac{1+y}{2}\right), & \zeta < 0, \quad y = (1 - 16\zeta)^{1/2}, \\ -5\zeta, & \zeta \geq 0. \end{cases} \quad (5.48)$$

Transfer coefficients.

$$C_D = \left[\frac{\kappa_{\text{vK}}}{\ln(z/z_0) - \psi_m(\zeta)} \right]^2, \quad C_H = \frac{\kappa_{\text{vK}}^2}{[\ln(z/z_0) - \psi_m(\zeta)][\ln(z/z_{0t}) - \psi_h(\zeta)]}. \quad (5.49)$$

The iterative solver (5 iterations via `jax.lax.fori_loop`) is fully differentiable.

5.7.3 COARE 3.0 Roughness

Over ocean ([Fairall et al., 2003](#)):

$$z_0 = \alpha_c \frac{u_*^2}{g} + 0.11 \frac{\nu}{u_*}, \quad z_{0t} = \min(1.15 \times 10^{-4}, 5.5 \times 10^{-5} \text{Re}_*^{-0.6}), \quad (5.50)$$

where $\alpha_c = 0.011$ is the Charnock parameter.

5.7.4 Large & Yeager (2004) Roughness

$$C_{DN} \times 10^3 = \frac{2.7}{U_{10N}} + 0.142 + 0.0764 U_{10N}. \quad (5.51)$$

5.8 Gravity Wave Drag

5.8.1 Orographic GWD (Lindzen 1981)

Surface stress.

$$\tau_0 = \rho N k h_{\text{topo}}^2 U, \quad N = \sqrt{\frac{g}{\theta} \frac{d\theta}{dz}}. \quad (5.52)$$

Saturation criterion. Waves break when the Froude number $\text{Fr} = U/(N h_{\text{topo}})$ drops below a critical value. Stress is capped at the saturation value $\tau_{\text{sat}} = \rho |U|^3 k/N$.

Momentum deposition.

$$\left. \frac{du}{dt} \right|_{\text{GWD}} = -\frac{1}{\rho} \frac{d\tau}{dz}. \quad (5.53)$$

Frictional heating.

$$\left. \frac{dT}{dt} \right|_{\text{GWD}} = -\frac{1}{c_p} \left(u \frac{du}{dt} + v \frac{dv}{dt} \right). \quad (5.54)$$

5.9 Held–Suarez Forcing

The idealised forcing benchmark of [Held and Suarez \(1994\)](#):

Equilibrium temperature.

$$T_{\text{eq}} = \max \left(200, \left[315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi \right] (p/p_0)^\kappa \right), \quad (5.55)$$

with $\Delta T_y = 60$ K, $\Delta \theta_z = 10$ K.

Newtonian relaxation.

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max \left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b} \right) \cos^4 \phi, \quad (5.56)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction.

$$\frac{d\mathbf{v}}{dt} = -k_v(\sigma) \mathbf{v}, \quad k_v = k_f \max \left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b} \right), \quad k_f = 1/(1 \text{ day}). \quad (5.57)$$

Chapter 6

Ocean Model

6.1 Boussinesq Hydrostatic Primitive Equations

In vector-invariant form on the cubed sphere:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (6.1)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (6.2)$$

The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies.

6.1.1 Tracer Equations (Skew-Symmetric)

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (6.3)$$

6.1.2 Free-Surface Equation

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (6.4)$$

6.2 Barotropic Solver

Split-explicit forward-backward substeps:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (6.5)$$

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (6.6)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$.

6.3 Equation of State (Wright 1997)

The ocean prognostic temperature variable T is **potential temperature** [degC]. The Wright (1997) EOS computes in-situ density $\rho(\theta, S, p)$ directly from potential temperature θ , salinity S [PSU], and pressure p [Pa]:

$$\rho = \frac{p + p_0(\theta, S)}{\lambda(\theta, S) + \alpha_0(\theta, S)(p + p_0)}, \quad (6.7)$$

where

$$\alpha_0 = a_0 + a_1\theta + a_2S, \quad (6.8)$$

$$p_0 = (b_0 + b_4S) + \theta(b_1 + \theta(b_2 + b_3\theta) + b_5S), \quad (6.9)$$

$$\lambda = (c_0 + c_4S) + \theta(c_1 + \theta(c_2 + c_3\theta) + c_5S). \quad (6.10)$$

Coefficients follow the MOM6 implementation. The density perturbation $\rho' = \rho(\theta, S, p) - \rho_0$ is used in the pressure gradient computation.

6.4 Hydrostatic Pressure and Vertical Velocity

Baroclinic pressure by downward integration:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (6.11)$$

Vertical velocity diagnosed bottom-up from $w(\text{floor}) = 0$:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (6.12)$$

6.5 Vertical Mixing: K-Profile Parameterisation (KPP)

Following [Large et al. \(1994\)](#):

6.5.1 Boundary Layer Depth

The boundary layer depth h is diagnosed where the bulk Richardson number crosses $\text{Ri}_{\text{crit}} = 0.25$ (LMD94 default):

$$\text{Ri}_b(z) = \frac{g \Delta \rho z}{\rho_0 (|\Delta \mathbf{V}|^2 + V_t^2)}, \quad (6.13)$$

using sigmoid-weighted averaging for a differentiable diagnosis.

6.5.2 K-Profile Shape

$$K_m(\sigma) = h w_s(\sigma) G(\sigma), \quad G(\sigma) = \sigma(1 - \sigma)^2, \quad \sigma = z/h, \quad (6.14)$$

where w_s is the turbulent velocity scale depending on surface fluxes and stability.

6.5.3 Non-Local Transport

For unstable (convective) boundary layers ($B_f > 0$), a non-local counter-gradient flux is added to temperature and salinity tendencies. The implementation uses a simplified form inspired by LMD94 Eq. 19:

$$\left. \frac{\partial T}{\partial t} \right|_{\text{NL}} = \gamma_T \frac{w_s(0) G(\sigma)}{w_s(\sigma) h} \left. \frac{\partial T}{\partial z} \right|_{\text{sfc}}, \quad (6.15)$$

where $\gamma_T = \gamma_S = 6.33$ are non-local transport coefficients and $w_s(0) (\partial T / \partial z)|_{\text{sfc}}$ is a surface heat flux proxy [K m/s]. This differs from the full LMD94 non-local term, which uses the explicit surface buoyancy flux $\overline{w'\theta'}_0$; the present form approximates it via the near-surface gradient and surface turbulent velocity scale. This is active only within the diagnosed boundary layer ($\sigma < 1$) and only under unstable forcing.

6.6 Lateral Mixing

6.6.1 Harmonic (Laplacian) Diffusion

$$D_h = A_h \nabla^2 \phi. \quad (6.16)$$

6.6.2 Biharmonic (∇^4) Diffusion

$$D_h = -A_4 \nabla^4 \phi. \quad (6.17)$$

6.6.3 Gent–McWilliams / Redi Parameterisation

The GM parameterisation represents the effect of mesoscale eddies on tracer transport via an additional advective flux (bolus velocity) (Gent and McWilliams, 1990):

$$\mathbf{u}^* = -\kappa_{\text{GM}} \nabla_h s, \quad (6.18)$$

where $s = -\nabla_h \rho / (\partial \rho / \partial z)$ is the isopycnal slope. The Redi diffusion tensor rotates lateral mixing along isopycnal surfaces.

6.7 Slab and Two-Layer Ocean

For idealised experiments, a simple mixed-layer energy balance:

$$C_{\text{mix}} \frac{dT_s}{dt} = \text{SW}_{\text{net}} + \text{LW}_{\text{net}} - \text{SH} - \text{LH} + Q_{\text{flux}} + K_{\text{mix}}(T_{\text{deep}} - T_s), \quad (6.19)$$

where $C_{\text{mix}} = \rho_{\text{ocn}} c_{\text{ocn}} h_{\text{mix}}$ is the mixed-layer heat capacity.

Chapter 7

Ocean Biogeochemistry

7.1 Carbonate Chemistry and Air–Sea CO₂ Exchange

7.1.1 Gas Transfer Velocity

Following Wanninkhof (2014):

$$k_w = 0.251 U_{10}^2 \left(\frac{\text{Sc}}{660} \right)^{-1/2}, \quad (7.1)$$

where the Schmidt number is a polynomial function of temperature:

$$\text{Sc} = 2116.8 - 136.25 T + 4.7353 T^2 - 0.092307 T^3 + 7.555 \times 10^{-4} T^4. \quad (7.2)$$

7.1.2 Net Air–Sea Flux

$$F_{\text{CO}_2} = k_w K_0 (p\text{CO}_{2,\text{ocean}} - p\text{CO}_{2,\text{atm}}), \quad (7.3)$$

where K_0 is Henry’s law solubility. The ocean surface $p\text{CO}_2$ is computed from DIC, ALK, T , S via carbonate equilibria using Newton–Raphson iteration.

7.2 NPZD Ecosystem Model

Four biological tracers: NO₃ (nutrients), P (phytoplankton), Z (zooplankton), D (detritus), following Fasham et al. (1990):

$$\frac{\partial P}{\partial t} = \mu_P P - G - m_P P, \quad (7.4)$$

$$\frac{\partial Z}{\partial t} = \gamma_Z G - m_Z Z^2, \quad (7.5)$$

$$\frac{\partial D}{\partial t} = (1 - \gamma_Z)G + m_P P + m_Z Z^2 - r_D D + w_s \frac{\partial D}{\partial z}, \quad (7.6)$$

$$\frac{\partial \text{NO}_3}{\partial t} = -\mu_P P + r_D D. \quad (7.7)$$

Phytoplankton growth.

$$\mu_P = \mu_{\max} f(T) \min(L_N, L_I), \quad L_N = \frac{\text{NO}_3}{k_N + \text{NO}_3}, \quad L_I = 1 - e^{-k_{\text{PAR}} \cdot \text{PAR}}. \quad (7.8)$$

Grazing (Holling III).

$$G = g_{\max} \frac{P^2}{K_g + P^2} Z. \quad (7.9)$$

Redfield stoichiometry ($R_{C:N} = 6.625$) couples biology to DIC/ALK.

Chapter 8

Land Surface Model

8.1 Slab Land

8.1.1 Surface Energy Balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) SW^\downarrow + \varepsilon_l (LW^\downarrow - \sigma_{SB} T_s^4) - SH - LH. \quad (8.1)$$

8.1.2 Bucket Hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\max}]. \quad (8.2)$$

Moisture availability $\beta = \beta_{\min} + (1 - \beta_{\min}) W/W_{\max}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

8.2 Multi-Layer Land Model

8.2.1 Soil Thermal Diffusion

$$C_{\text{eff}}(z) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[k_{\text{eff}}(z) \frac{\partial T}{\partial z} \right], \quad (8.3)$$

with effective heat capacity:

$$C_{\text{eff}} = (1 - \theta_{\text{sat}}) C_{\text{soil}} + \theta C_{\text{water}} + (\theta_{\text{sat}} - \theta) C_{\text{air}}, \quad (8.4)$$

and thermal conductivity following the [Johansen \(1975\)](#) parameterisation via the Kersten number. Solved implicitly using the Thomas algorithm with surface heat flux and geothermal flux boundary conditions.

8.2.2 Richards Equation (Soil Hydraulics)

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right], \quad (8.5)$$

where θ is volumetric water content, ψ is matric potential, and $K(\psi)$ is hydraulic conductivity.

Six water retention curves are available:

Curve	$\psi(\theta)$
Van Genuchten (1980)	$\psi = \alpha^{-1}[(S_e^{-1/m} - 1)^{1/n}]$
Campbell (1974)	$\psi = \psi_s(S_e)^{-b}$
Brooks–Corey (1964)	$\psi = \psi_d(S_e)^{-1/\lambda}$
Clapp–Hornberger (1978)	Power-law form

where $S_e = (\theta - \theta_r)/(\theta_{\text{sat}} - \theta_r)$ is effective saturation.

Root water uptake. Root distribution: $f_{\text{root}}(z) = \exp(-z/z_{\text{root}})$, with default $z_{\text{root}} = 0.5$ m. Plant-available water between wilting point (θ_{wp}) and field capacity (θ_{fc}).

8.2.3 Snow Budget

$$\frac{dh_{\text{swe}}}{dt} = P_{\text{snow}} - M, \quad M = \max\left(0, \frac{Q_{\text{net}}}{L_f \rho_w}\right), \quad (8.6)$$

with age-dependent albedo: $\alpha_{\text{snow}}(t) = \alpha_{\text{max}} e^{-t/\tau_{\text{age}}} + \alpha_{\text{min}}(1 - e^{-t/\tau_{\text{age}}})$.

8.3 Land Carbon Cycle

8.3.1 DALEC-990 Six-Pool Model

Six prognostic pools: labile C_{lab} , foliage C_f , root C_r , wood C_w , litter C_{lit} , and soil organic matter C_{SOM} .

Gross Primary Production (GPP).

$$\text{GPP} = \varepsilon \text{APAR} f_T f_{\text{CO}_2} \beta, \quad \text{APAR} = f_{\text{APAR}} \text{PAR}, \quad f_{\text{APAR}} = 1 - e^{-k_{\text{ext}} \text{LAI}}, \quad (8.7)$$

where ε is light-use efficiency, $f_T = \exp[-0.5((T - T_{\text{opt}})/T_{\text{width}})^2]$ is the Gaussian temperature response, and $f_{\text{CO}_2} = \text{CO}_2/(K_{\text{CO}_2} + \text{CO}_2)$ is Michaelis–Menten CO_2 response.

Soil respiration. Exponential Q_{10} temperature sensitivity with moisture limitation.

8.3.2 Farquhar (1980) Photosynthesis

Assimilation rate:

$$A = \min(W_c, W_j) - R_d, \quad (8.8)$$

where:

$$W_c = V_{c,\text{max}} \frac{C_i - \Gamma^*}{C_i + K_c(1 + O/K_o)}, \quad (\text{Rubisco-limited}), \quad (8.9)$$

$$W_j = \frac{J(C_i - \Gamma^*)}{4(C_i + 2\Gamma^*)}, \quad (\text{light-limited}). \quad (8.10)$$

$V_{c,\text{max}}$ and J_{max} follow peaked Arrhenius temperature responses (Bernacchi et al., 2001). The minimum is implemented as a smooth (differentiable) function.

8.3.3 Stomatal Conductance

Three models couple stomata to photosynthesis:

Ball–Berry (1987). $g_s = g_0 + g_1 A \text{RH}/C_s$.

Medlyn (2011). $g_s = g_0 + 1.6(1 + g_1/\sqrt{\text{VPD}}) A/C_s$.

Jarvis (1976). $g_s = g_{s,\max} f(\text{PAR}) f(T) f(\text{VPD}) f(\theta)$ (CO₂-independent; for when the carbon cycle is inactive).

The coupled A – g_s – C_i system is solved by 5 fixed-point iterations, unrolled for JAX JIT compatibility.

Chapter 9

Sea Ice

9.1 Thermodynamic Slab

9.1.1 Surface Temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i(T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\text{min}}). \quad (9.1)$$

9.1.2 Growth and Melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (9.2)$$

9.1.3 Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1 - c)/h_{\text{new}} & \text{growing,} \\ (dh/dt) c/h_{\text{eff}} & \text{melting.} \end{cases} \quad (9.3)$$

9.2 EVP Rheology

Following [Hunke and Dukowicz \(1997\)](#), the ice momentum equation is

$$m \frac{\partial \mathbf{u}_i}{\partial t} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_w - m f \hat{\mathbf{k}} \times \mathbf{u}_i - mg \nabla \eta + \nabla \cdot \boldsymbol{\sigma}, \quad (9.4)$$

where $\boldsymbol{\sigma}$ is the internal stress tensor from an elliptic yield curve with eccentricity $e = 2$. Ice strength:

$$P = P^* h \exp(-C(1 - c)), \quad P^* = 27.5 \text{ kPa}, \quad C = 20. \quad (9.5)$$

The elastic term provides subcycling stability ($N_{\text{evp}} = 120$ substeps).

9.3 Multi-Category Ice Thickness Distribution

For $N_{\text{cat}} > 1$, per-category state (h_k, T_k, c_k) evolves independently under thermodynamics, then [Lipscomb \(2001\)](#) linear remapping redistributes ice across thickness categories, maintaining a well-resolved thickness distribution.

Chapter 10

Tile-Based Coupler

10.1 Surface Flux Blending

Fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (10.1)$$

Tile fractions satisfy $\sum f_i = 1$ with consistency checks.

10.2 Ocean Albedo

The Briegleb (1992) zenith-angle-dependent formula:

$$\alpha_{\text{ocn}} = \frac{0.026}{1.1 \mu_0^{1.7} + 0.065} + 0.15 (\mu_0 - 0.1)(\mu_0 - 0.5)(\mu_0 - 1.0), \quad (10.2)$$

where $\mu_0 = \cos \theta_z$.

10.3 Lake Model

A two-layer model with epilimnion (warm, mixed) and hypolimnion (cold, deep) connected by entrainment mixing.

Chapter 11

Machine Learning Components

11.1 Spherical Fourier Neural Operator (SFNO)

11.1.1 Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. N **processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** output = input + decoder($\dots$).

11.1.2 Spectral Convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (11.1)$$

where $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair.

The SFNO can serve as a learned dynamical core or physics emulator within the legoESM framework, sharing the same spectral transform infrastructure as the conventional spectral cores.

Chapter 12

Data Assimilation

12.1 Incremental 4D-Var

legoESM provides a fully implemented incremental 4D-Var data assimilation system, leveraging JAX’s automatic differentiation to compute exact adjoints without hand-coded tangent-linear or adjoint models.

12.1.1 Control Vector

The control vector $\mathbf{x}$ comprises the prognostic atmospheric fields:

- Horizontal winds (u, v) ;
- Temperature T ;
- Specific humidity q ;
- Surface pressure p_s .

The control vector specification supports both grid-point and spectral representations, with configurable variable subsets for targeted assimilation.

12.1.2 Cost Function

The cost function minimises the sum of background and observation terms:

$$J(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x} - \mathbf{x}_b) + \frac{1}{2} \sum_i (\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i)^T \mathbf{R}_i^{-1}(\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i), \quad (12.1)$$

where $\mathbf{x}_b$ is the background state, $\mathbf{B}$ is the background error covariance, $\mathcal{M}$ is the forward model, $\mathcal{H}_i$ are observation operators, $\mathbf{y}_i$ are observations, and $\mathbf{R}_i$ are observation error covariances.

12.1.3 Background Error Covariance

Four $\mathbf{B}$ matrix families:

- **Diagonal:** $\mathbf{B} = \text{diag}(\sigma_b^2)$;
- **Diffusion:** Implicit diffusion-based correlation modelling;
- **Spectral:** Diagonal in spectral space (scale-dependent variances);
- **Hybrid:** Linear combination of flow-dependent ensemble covariance and static $\mathbf{B}$.

12.1.4 Minimisation

L-BFGS and conjugate gradient solvers with optional $\mathbf{B}^{1/2}$ preconditioning for improved convergence. The gradient $\nabla_{\mathbf{x}} J$ is computed exactly via `jax.grad` through the full forward model trajectory.

12.1.5 Observation Operators

Three observation operator types:

- **Direct:** Identity mapping for in-situ observations on model grid points;
- **Interpolating:** Bilinear/trilinear interpolation from model grid to observation locations;
- **Composite:** Chained operators for derived quantities (e.g., satellite radiances via a forward radiative transfer step).

All operators are JAX-differentiable, so `jax.grad` propagates through observation-space transformations seamlessly.

12.1.6 Analysis–Forecast Cycling

Sequential cycling: the analysis from one assimilation window becomes the background for the next, with checkpoint/restart support.

12.1.7 Incremental Analysis Update

To reduce initialisation shocks, the analysis increment $\delta\mathbf{x} = \mathbf{x}_a - \mathbf{x}_b$ is applied gradually over an IAU window of N_{IAU} timesteps:

$$\mathbf{x}^{n+1} = \mathcal{M}(\mathbf{x}^n) + \frac{\delta\mathbf{x}}{N_{\text{IAU}}}, \quad n = 0, \dots, N_{\text{IAU}} - 1. \quad (12.2)$$

This produces dynamically balanced analyses and is the default cycling mode.

Chapter 13

Conservation and Numerical Corrections

13.1 Atmospheric Mass Conservation

Dry-air mass is corrected by a uniform adjustment to p_s :

$$\Delta\bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - M_{\text{target}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta\bar{p}. \quad (13.1)$$

Initial-mass anchoring. For long integrations, $M_{\text{target}} = M_0 = \sum_i A_i p_{s,i}^{\text{init}}$ prevents cumulative $O(\varepsilon)$ drift. All accumulations use float64.

Moisture conservation. Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$.

13.2 Ocean Conservation

Volume: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt: uniform additive correction per unit volume.

Chapter 14

Differentiable Training

The `training/` module provides end-to-end differentiable training pipelines for three modes:

14.1 Physics Parameter Tuning

`TrainablePhysicsParams` wraps 8 physics parameters (convective relaxation time, mixing length, drag coefficients, etc.) as an Equinox module with sigmoid-based constraints mapping from unconstrained space to valid physical ranges. Gradients flow through `eqx.filter_value_and_grad` applied to the model's `build_segment_fn(...).raw` (non-JIT, non-donating variant for AD compatibility).

14.2 Neural GCM Training

Spectral neural GCM training couples a learned column model with the spectral dynamical core. The loss function combines area-weighted MSE, spectral loss (penalising errors at each total wavenumber), and per-variable MSE, imported from `ml/loss.py`. Optimisation uses `ml/training.create_optimizer` (warmup + cosine decay + gradient clipping).

14.3 SFNO–Dycore Coupling

The Spherical Fourier Neural Operator can be coupled to any dynamical core for hybrid learned–physical simulations. Training uses the same segment-based architecture, with `SegmentForcing` as an explicit argument (not closure-captured) to prevent JIT recompilation when forcing data changes.

14.4 ERA5 Data Ingestion

`era5.to_state.py` converts ERA5 reanalysis (lat-lon) to any model grid with a local Zarr cache for efficient repeated access. Vertical interpolation handles pressure-to-sigma/hybrid coordinate mapping.

Chapter 15

Parallel Runtime

15.1 Execution Modes

Mode	Ranks	Devices/Rank	Halo Backend	Reduction Backend
serial	1	1	local	local
multi_device	1	N	JAX SPMD	local
mpi	N	1	MPI	MPI
hybrid	N	M	MPI + JAX	MPI + JAX

The `ParallelRuntime.create()` entry point detects MPI rank/size, validates device counts, and routes to the correct halo/reduction backend.

15.2 Domain Decomposition

Cubed sphere. Valid device counts: 1, 2, 3, 6, and $6k^2$ for $k \geq 2$. Each face can be further tiled into sub-face patches. `initialize_distributed(global_n=N) → scatter_to_local()` distributes the global state to rank-local patches; `gather_to_global()` is used only for I/O.

Voronoi/MPAS. RCB geometric partitioner (no external dependencies) or METIS graph partitioner. Two-layer halo for biharmonic operators.

15.3 MPI Differentiability

All halo exchange `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Global reductions use AD-safe `allreduce(SUM)` with a custom VJP; `MAX`, `MIN`, `allgather`, and `bcast` are not differentiable and are used only in diagnostics. Conservation fixers (`fix_mass`, `zero_mean_tendency`) correctly flow gradients through global reductions.

The native 4D halo exchange path (`pad_halo_4d`, `pad_halo_vector_4d`) exchanges all vertical levels in one MPI message per neighbour, replacing the earlier per-level `vmap(pad_halo)` which issued n_{lev} separate messages.

15.4 Ensemble Parallelism

`jax.vmap` across ensemble members with `jax.lax.scan` over time steps. Support for perturbed initial conditions, ensemble mean/std/percentile diagnostics, and gradient checkpointing.

Chapter 16

CMIP Infrastructure

16.1 CF/CMOR-Compliant Output

CF-1.8 NetCDF4 files with CMIP6 Data Reference Syntax (DRS) filenames: `<var>_<table>_<model>_<expt>_<var>`
Four CMOR tables: Amon (33 atmospheric variables including clear-sky radiative fluxes, cloud condensate, and relative humidity), Lmon (6 land variables), Omon (2 ocean variables: SST and sea-ice fraction), and Aday (5 daily atmospheric variables) on a standard 19-level pressure grid.

16.2 Experiment Templates

Template	Period	Forcing	CO ₂ range
piControl	1850– ∞	Fixed 1850	284.3 ppmv
historical	1850–2014	Transient	284→397 ppmv
ssp245	2015–2100	SSP2-4.5	397→600 ppmv
ssp585	2015–2100	SSP5-8.5	397→1135 ppmv
amip	1979–2014	Prescribed SST	Transient
1pctCO2	1–150	1%/yr increase	284→1277 ppmv

16.3 Restart and Reproducibility

Each restart file includes a `.meta.json` sidecar with SHA-256 state digest, config hash, platform identifier, JAX version, x64 mode, and git hash.

Chapter 17

Verification and Testing

The legoESM codebase comprises approximately 385 source modules ($\sim 102,000$ lines of code), validated by 3,489 tests across 216 test files.

17.1 Williamson Standard Test Cases

The shallow water dynamical cores are verified against the standard Williamson test suite (Williamson et al., 1992), including:

- **Case 1:** Advection of a cosine bell;
- **Case 2:** Global steady-state nonlinear geostrophic flow;
- **Case 5:** Zonal flow over an isolated mountain;
- **Case 6:** Rossby–Haurwitz wave.

A comprehensive diagnostic suite (`williamson_diagnostic.py`) computes L_1 , L_2 , and L_∞ error norms against analytical solutions where available, and performs cross-discretisation comparison across all five grid families (cubed-sphere C-D, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi).

17.2 Spectral Dynamical Core Test Suite

The spectral transform dynamical cores (shallow water and primitive equation) are validated by a dedicated comprehensive test suite covering:

- **Conservation:** Total mass, energy, and enstrophy conservation to machine precision (shallow water) and within bounded drift (primitive equation with hyperdiffusion);
- **Stability:** Long-integration stability under Held–Suarez forcing at T21/L40 and T42/L40 resolutions;
- **Spectral convergence:** Verification that error norms decrease with increasing truncation wavenumber;
- **Semi-implicit correctness:** Validation that the implicit gravity-wave solver matches reference solutions and permits CFL numbers > 1 for gravity-wave modes.

17.3 Cross-Discretisation Intercomparison

All five shallow water discretisations are compared on the same test problems, verifying consistent convergence rates and inter-scheme agreement in error norms. This ensures that the shared physics

interface produces consistent results regardless of the dynamical core choice.

17.4 MPI Distributed Tests

Dedicated distributed test suites (`tests/distributed/`) verify:

- **Halo exchange correctness:** Scalar and vector 4D halo exchange on 2–6 ranks matches serial reference within round-off;
- **MPI differentiability:** `jax.grad` through halo-exchange and allreduce operations produces gradients matching single-rank reference (`test_mpi_differentiability.py`);
- **MPI driver integration:** Full model stepping under MPI produces results consistent with serial execution (`test_mpi_driver.py`).

Chapter 18

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth mean radius	6.371×10^6	m
R_d	Dry-air gas constant	287.05	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air specific heat (c_p)	1004.64	$\text{J kg}^{-1} \text{K}^{-1}$
c_v	Dry-air specific heat (c_v)	717.56	$\text{J kg}^{-1} \text{K}^{-1}$
κ	Poisson constant R_d/c_p	0.2857	—
R_v	Water-vapour gas constant	461.51	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pv}	Specific heat of water vapour	1846	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pw}	Specific heat of liquid water	4218	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pi}	Specific heat of ice	2106	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
L_s	Latent heat of sublimation	2.834×10^6	J kg^{-1}
L_f	Latent heat of fusion	3.337×10^5	J kg^{-1}
ε	R_d/R_v	0.622	—
T_{freeze}	Freezing point of water	273.15	K
ρ_w	Liquid water density	1000	kg m^{-3}
ρ_i	Ice density	917	kg m^{-3}
p_0	Reference pressure	10^5	Pa
S_0	Total solar irradiance	1361	W m^{-2}
σ_{SB}	Stefan–Boltzmann constant	5.670×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
ρ_0	Ocean reference density	1025	kg m^{-3}

References

- Audusse, E., F. Bouchut, M.-O. Bristeau, R. Klein, and B. Perthame, 2004: A fast and stable well-balanced scheme with hydrostatic reconstruction for shallow water flows. *SIAM J. Sci. Comput.*, **25**, 2050–2065.
- Ball, J.T., I.E. Woodrow, and J.A. Berry, 1987: A model predicting stomatal conductance and its contribution to the control of photosynthesis under different environmental conditions. *Prog. in Photosynthesis Res.*, **4**, 221–224.
- Bernacchi, C.J., E.L. Singaas, C. Pimentel, A.R. Portis, and S.P. Long, 2001: Improved temperature response functions for models of Rubisco-limited photosynthesis. *Plant Cell Environ.*, **24**, 253–259.
- Bourke, W., 1972: An efficient, one-level, primitive-equation spectral model. *Mon. Wea. Rev.*, **100**, 683–689.
- Businger, J.A., J.C. Wyngaard, Y. Izumi, and E.F. Bradley, 1971: Flux-profile relationships in the atmospheric surface layer. *J. Atmos. Sci.*, **28**, 181–189.
- Colella, P. and P.R. Woodward, 1984: The Piecewise Parabolic Method (PPM) for gas-dynamical simulations. *J. Comput. Phys.*, **54**, 174–201.
- Fairall, C.W., E.F. Bradley, J.E. Hare, A.A. Grachev, and J.B. Edson, 2003: Bulk parameterization of air–sea fluxes: Updates and verification for the COARE algorithm. *J. Climate*, **16**, 571–591.
- Farquhar, G.D., S. von Caemmerer, and J.A. Berry, 1980: A biochemical model of photosynthetic CO₂ assimilation in leaves of C3 species. *Planta*, **149**, 78–90.
- Fasham, M.J.R., H.W. Ducklow, and S.M. McKelvie, 1990: A nitrogen-based model of plankton dynamics in the oceanic mixed layer. *J. Mar. Res.*, **48**, 591–639.
- Frierson, D.M.W., I.M. Held, and P. Zurita-Gotor, 2006: A gray-radiation aquaplanet moist GCM. Part I: Static stability and eddy scale. *J. Atmos. Sci.*, **63**, 2548–2566.
- Frierson, D.M.W., 2007: The dynamics of idealized convection schemes and their effect on the zonally averaged tropical circulation. *J. Atmos. Sci.*, **64**, 1959–1976.
- Gent, P.R. and J.C. McWilliams, 1990: Isopycnal mixing in ocean circulation models. *J. Phys. Oceanogr.*, **20**, 150–155.
- Held, I.M. and M.J. Suarez, 1994: A proposal for the intercomparison of the dynamical cores of atmospheric general circulation models. *Bull. Amer. Meteor. Soc.*, **75**, 1825–1830.

- Holtstlag, A.A.M. and B.A. Boville, 1993: Local versus nonlocal boundary-layer diffusion in a global climate model. *J. Climate*, **6**, 1825–1842.
- Hoskins, B.J. and A.J. Simmons, 1975: A multi-layer spectral model and the semi-implicit method. *Quart. J. Roy. Meteor. Soc.*, **101**, 637–655.
- Hunke, E.C. and J.K. Dukowicz, 1997: An elastic-viscous-plastic model for sea ice dynamics. *J. Phys. Oceanogr.*, **27**, 1849–1867.
- Jiang, G.-S. and C.-W. Shu, 1996: Efficient implementation of weighted ENO schemes. *J. Comput. Phys.*, **126**, 202–228.
- Johansen, O., 1975: Thermal conductivity of soils. Ph.D. thesis, University of Trondheim, 236 pp.
- Kessler, E., 1969: On the Distribution and Continuity of Water Substance in Atmospheric Circulations. *Meteor. Monogr.*, No. 32, Amer. Meteor. Soc., 84 pp.
- Large, W.G., J.C. McWilliams, and S.C. Doney, 1994: Oceanic vertical mixing: A review and a model with a nonlocal boundary layer parameterization. *Rev. Geophys.*, **32**, 363–403.
- Large, W.G. and S.G. Yeager, 2004: Diurnal to decadal global forcing for ocean and sea-ice models: The data sets and flux climatologies. *NCAR Technical Note*, NCAR/TN-460+STR.
- Lin, S.-J., 2004: A “vertically Lagrangian” finite-volume dynamical core for global models. *Mon. Wea. Rev.*, **132**, 2293–2307.
- Lipscomb, W.H., 2001: Remapping the thickness distribution in sea ice models. *J. Geophys. Res.*, **106**(C7), 13989–14000.
- Louis, J.-F., 1979: A parametric model of vertical eddy fluxes in the atmosphere. *Bound.-Layer Meteor.*, **17**, 187–202.
- Medlyn, B.E., et al., 2011: Reconciling the optimal and empirical approaches to modelling stomatal conductance. *Global Change Biol.*, **17**, 2134–2144.
- Morrison, H., J.A. Curry, and V.I. Khvorostyanov, 2005: A new double-moment microphysics parameterization for application in cloud and climate models. Part I: Description. *J. Atmos. Sci.*, **62**, 1665–1677.
- Pincus, R., E.J. Mlawer, and J.S. Delamere, 2019: Balancing accuracy, efficiency, and flexibility in radiation calculations for dynamical models. *J. Adv. Model. Earth Syst.*, **11**, 3074–3089.
- Mouallem, J., L.M. Harris, and R. Benson, 2023: A duo-grid methodology for cubed-sphere shallow water models. *J. Comput. Phys.*, **485**, 112097.
- Putman, W.M. and S.-J. Lin, 2007: Finite-volume transport on various cubed-sphere grids. *J. Comput. Phys.*, **227**, 55–78.
- Ringler, T.D., J. Thuburn, J.B. Klemp, and W.C. Skamarock, 2010: A unified approach to energy conservation and potential vorticity dynamics for arbitrarily-structured C-grids. *J. Comput. Phys.*, **229**, 3065–3090.
- Seifert, A. and K.D. Beheng, 2001: A two-moment cloud microphysics parameterization for mixed-phase clouds. *Meteor. Atmos. Phys.*, **73**, 67–88.

- Shu, C.-W. and S. Osher, 1988: Efficient implementation of essentially non-oscillatory shock-capturing schemes. *J. Comput. Phys.*, **77**, 439–471.
- Simmons, A.J. and D.M. Burridge, 1981: An energy and angular-momentum conserving vertical finite-difference scheme and hybrid vertical coordinates. *Mon. Wea. Rev.*, **109**, 758–766.
- Skamarock, W.C. and J.B. Klemp, 2008: A time-split nonhydrostatic atmospheric model for weather research and forecasting applications. *J. Comput. Phys.*, **227**, 3465–3485.
- Van Genuchten, M.Th., 1980: A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.*, **44**, 892–898.
- van Leer, B., 1979: Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov’s method. *J. Comput. Phys.*, **32**, 101–136.
- Wanninkhof, R., 2014: Relationship between wind speed and gas exchange over the ocean revisited. *Limnol. Oceanogr. Methods*, **12**, 351–362.
- Williams, P.D., 2009: A proposed modification to the Robert–Asselin time filter. *Mon. Wea. Rev.*, **137**, 2538–2546.
- Williamson, D.L., J.B. Drake, J.J. Hack, R. Jakob, and P.N. Swarztrauber, 1992: A standard test set for numerical approximations to the shallow water equations in spherical geometry. *J. Comput. Phys.*, **102**, 211–224.
- Wright, D.G., 1997: An equation of state for use in ocean models: Eckart’s formula revisited. *J. Atmos. Ocean. Tech.*, **14**, 735–740.